

Multidimensional Balancing

How DataAI builds starting matrices, target matrices, balancing coefficients, and AI-ready comparison outputs

Based on: MultidimensionalBalancing.aspx, MultidimensionalBalancing.aspx.vb, and the balancing helpers in App_Code/mFunctions.vb.

Important distinction: the page contains both classic matrix balancing and true multidimensional balancing. Scenarios 1a-3c balance two-dimensional row/column matrices or wide matrices where selected fields act as columns. Only scenarios 4a, 4b, and 4c are true multidimensional balancing, because they balance sums by multiple selected dimensions/columns rather than by one row field and one column field.

Multidimensional balancing is used when a starting data structure must be adjusted so its selected-dimensional totals match a target structure while preserving as much of the starting pattern as possible. In business terms, it answers: if the totals changed across several dimensions, what cell-level changes are consistent with the original data and the requested target totals?

What The Page Does

The page starts from the current report data already loaded into Session("dv3"). The user selects categories, measures, aggregation functions, scenario, iteration settings, and optional manual target sums. The code converts those selections into one or more matrices or multidimensional summary tables, balances the starting structure to the target totals, and displays the resulting matrices, coefficients, comparisons, exports, and AI interpretation links.

- **Starting Matrix:** the original matrix or multidimensional table built from report data. In scenarios 1a-3c it is primarily a row/column matrix. In scenarios 4a-4c it is built by selected dimensions.
- **Target Matrix or Target Sums:** the requested or actual target structure. It may be entered manually, derived from Field2, or derived from the same Field1 under a different Field2 value such as year, period, scenario, or status.
- **Balancing Matrix:** the adjusted structure. It has totals close to the target totals, while keeping the starting proportions as much as the iterative method allows.
- **Balancing Coefficients:** multipliers applied during iterations. These are the main explanation of how the starting data had to change to fit the target.
- **Precision and Number of Steps:** stopping controls. The method stops when the total mismatch is below Precision or when Number of Steps is reached.

Main Inputs And Controls

Control	What it means	Used in
Select Scenario	Chooses one of the balancing models. 1a-3c are row/column or wide matrix balancing. 4a-4c are true multidimensional balancing.	All scenarios
Row field	Defines the matrix rows for classic two-dimensional scenarios.	1a, 2a, 2b, 2c, 3a; row side of 1b, 3b, 3c
Column field	Defines the matrix columns for classic two-dimensional scenarios.	1a, 2a, 2b, 2c, 3a
Matrix items by Field1 + aggregation	Measure used for the starting matrix or multidimensional starting values. Numeric fields can use Sum, Avg, Min, Max, StDev; text fields use Count or CountDistinct.	Most scenarios
Field2 + aggregation	Measure used for target matrix values, or a field whose selected values define start/target filters.	2a, 2b, 2c, 4a, 4b
Starting value / Target value	Specific values of Field2 used to filter data and build starting and target matrices.	2b, 2c, 3b, 3c, 4b

Control	What it means	Used in
Multiple fields	Selected columns that either become matrix columns or become multiple target measures. This is not the same as true multidimensional balancing unless scenario 4a-4c is selected.	1b, 3a, 3b, 3c
Balance by selected columns	Selected dimensions used for true multidimensional sums rather than a simple row-by-column matrix.	4a, 4b, 4c
Sums by rows / Sums by columns	Manual target totals entered as comma-separated arrays. The page scales column sums when row and column totals differ.	1a, 1b
Number of Steps	Maximum number of balancing iterations.	All balancing buttons
Precision	Target mismatch threshold. Smaller precision means tighter balancing but may require more steps.	All balancing buttons
Adjust by starting values	When checked, balanced results can be scaled back to the starting total rather than target total.	Several scenarios

Algorithm In Plain Language

The core balancing helper uses an iterative proportional fitting / RAS-style process. The code calls BalanceMatrix through CanBalanceMatrixToColRowSums when target row and column sums are supplied, and through CanTargetMatrixBalancedFromStartingOne when a full target matrix is available. For true multidimensional scenarios 4a-4c, the page first converts selected dimensions into starting and target dimensional sums, then applies the balancing logic to reconcile those sums.

- Step 1: build the starting matrix or multidimensional table from report data using selected fields, dimensions, and aggregation functions.
- Step 2: build or read the target totals. Targets can come from manual sums, a target matrix, selected Field2 values, or multidimensional sums.
- Step 3: if totals differ, scale target totals or matrices so the starting and target totals can be compared on the same overall basis.
- Step 4: multiply rows/dimensions by coefficients so totals move toward target totals.
- Step 5: repeat coefficient adjustment across the other axis or dimensions so all target totals are approached together.
- Step 6: repeat until mismatch is below Precision or the maximum Number of Steps is reached.
- Step 7: return the balanced structure, coefficient table, precision message, and comparison coloring/tooltips.

Scenario Guide

Scenario	Type	Model	Typical use
1a	Two-dimensional matrix	Starting matrix = Field1 aggregated by row field and column field. Target = manually entered row sums and column sums.	Reallocate existing data to a requested budget, capacity, or control total.
1b	Wide matrix	Rows come from one group field; selected data fields become matrix columns. Target = manually entered row and column sums.	Balance a wide table where several numeric columns are the matrix columns.
2a	Two-dimensional matrix	Starting matrix from Field1. Target matrix from Field2. The balanced matrix matches target row/column totals.	Compare two measures with the same category structure, such as sales vs profit or actual vs forecast.

Scenario	Type	Model	Typical use
2b	Two-dimensional matrix over Field2 values	Starting and target matrices use Field1, filtered by starting and target values of Field2.	Compare one period/status/scenario to another, for example Year=2025 to Year=2026.
2c	Coefficient trend for 2D matrix	Runs coefficient analysis for all iterations between starting and target values of Field2.	See how coefficients evolve across ordered time, period, or scenario values.
3a	Multiple target matrices	Starting matrix from Field1; multiple selected fields become multiple target matrices.	Run the same starting matrix against several candidate target measures.
3b	Wide matrix over Field2 values	Rows are grouped; selected columns are matrix columns; starting/target Field2 values define the two matrices.	Compare wide-format matrices across two time periods or scenarios.
3c	Coefficient trend for wide matrix	Coefficient analysis for scenario 3b across all values between starting and target Field2 values.	Trend interpretation for a wide matrix through several periods.
4a	Multidimensional	Starting table from Field1; target multidimensional sums from Field2 by selected dimensions.	Balance by more than two category dimensions, not only rows and columns.
4b	Multidimensional over Field2 values	Starting and target multidimensional tables use Field1 filtered by starting/target Field2 values.	Compare multidimensional development between two values, such as two years.
4c	Multidimensional with manual targets	Starting multidimensional table from Field1; target sums are manually entered by selected dimensions.	Force data to requested dimensional control totals.

Sample Data And Example

The example below shows the simple scenario 1a pattern. This is two-dimensional matrix balancing, not true multidimensional balancing. Scenarios 4a-4c use the same principle, but selected dimensions replace the simple row/column structure.

Region	Channel	Sales
North	Online	120
North	Retail	80
South	Online	90
South	Retail	110

Starting Matrix

Region	Online	Retail	Total
North	120	80	200
South	90	110	200
Total	210	190	400

Requested target totals: row totals North=220, South=180; column totals Online=230, Retail=170. The grand total stays 400.

Balanced Matrix After Iterative Rescaling

Region	Online	Retail	Total
North	141.11	78.89	220.00
South	88.89	91.11	180.00
Total	230.00	170.00	400.00

Interpretation: North receives a larger share of the final matrix, and Online receives a larger column total. The balanced cells are not arbitrary: each cell changes according to row and column multipliers so the final table satisfies both row and column constraints.

How This Differs From True Multidimensional Scenarios

In scenarios 4a-4c, the user selects multiple dimensions in Balance by selected columns. The page builds sums for each selected dimension and balances the starting values against those dimensional target sums. Instead of asking only whether North and Online totals match, a true multidimensional scenario can ask whether Region, Product, Channel, Department, Location, or other selected dimensions all reconcile to their targets at the same time.

Example Coefficients

Step	ki North	ki South	kj Online	kj Retail	Precision meaning
1	1.10000	0.90000	1.07981	0.90909	First row pass then column pass
Final	varies by iteration	varies by iteration	varies by iteration	varies by iteration	Stops when mismatch < Precision or steps are exhausted

How To Use The Page

- Open the report data first so Session("dv3") is available; the page uses the already-loaded report data.

- Select a scenario. Use 1a-3c for row/column or wide matrix balancing. Use 4a-4c when the goal is true multidimensional balancing by selected dimensions.
- Choose row field, column field, Field1, Field2, aggregation functions, selected multiple fields, or selected dimensions as required by the scenario.
- Enter manual sums when using scenarios 1a, 1b, or 4c. For row/column manual sums, the number of values must match matrix dimensions. For 4c, manual sums correspond to selected multidimensional targets.
- Set Number of Steps and Precision. Start with 100 steps and precision 1, then tighten if the result is not close enough.
- Click the scenario button. The page builds matrices or dimensional sums, runs balancing, fills grids, applies color/tooltips, and stores grid data in Session for export or AI.
- Use Export links to download individual grids. Use AI links to send only the corresponding grid to ChatAI for interpretation.

Output Grids

Grid / section	Meaning
Starting Matrix	Original matrix or dimensional sums created from the selected report fields and filters. In 4a-4c, this represents multidimensional selected-column totals.
Target Matrix	Requested or actual target matrix/sums. In 4c this is manually entered target sums; in 2a/4a it comes from Field2; in 2b/3b/4b it comes from the target Field2 value.
Balancing Matrix	Calculated balanced expectation. It should match target row, column, or selected-dimensional sums within the requested precision.
Balancing coefficients	Iteration-level or final multipliers. In classic row/column balancing, ki columns are row weights and kj columns are column weights. In multidimensional balancing, the coefficient table also shows StartValue, TargetValue, BalancedValue, and DifferenceTargetBalanced or DifferenceStartBalanced.
Whole Matrix balancing	Shown when partial balancing parameters are used. It compares partial balancing against balancing the whole matrix. TextBoxUV is hidden by default and set to 0,0, so this area usually stays hidden.
Comparison label and colors	Cells are colored and tooltips show percent differences between starting, target, and balanced values. The result label reports precision and maximum difference where calculated.
Chart link	The coefficients grid can expose a chart link when matrix chart data is prepared.

Interpreting Results

- Small differences between Target Matrix and Balancing Matrix mean the balancing found a good fit.
- Large coefficient values identify rows, columns, or dimensions that had to move strongly to satisfy the target. These are often the best business explanation points.
- For 1a-3c, think in terms of row/column matrix movement. For 4a-4c, think in terms of selected-dimensional movement across several fields at once.
- DifferenceTargetBalanced shows where the balanced expectation still differs from the actual/requested target. DifferenceStartBalanced is used in manual-target scenario 4c, where the target values are not directly comparable row-by-row in the same way.
- If many cells are zero in the starting matrix, balancing may be less stable. The helper replaces zero cells with tiny positive values so the math can continue, but interpretation should be cautious.
- If the page reports dimensions do not match, the selected sums or target matrix do not match the row/column or selected-field structure of the starting matrix.
- If the page reaches the step limit without acceptable precision, increase Number of Steps, relax Precision, or check whether target totals are structurally inconsistent with the starting matrix.

AI And Export Workflow

Each output section has an AI link that prepares a specific question and sends only that grid data to ChatAI. Starting, target, balanced, coefficient, whole-matrix, and target-vs-balanced comparison outputs can be interpreted separately. This is useful because each grid answers a different analytical question: what changed, where the target pulls the structure, how the balanced expectation differs, and which coefficients explain the movement.

- Export to Excel links call the project export routine and download grid data from Temp as a delimited/CSV-style export file.
- AI links populate Session("QuestionToAI"), Session("dataTable"), and Session("DataToChatAI") with the selected grid only.
- The Target Sums <=> Balancing Sums AI link creates a differences table from target and balanced matrices using DifferencesOfTwoMatrices.
- The page does not require machine learning. It is deterministic matrix balancing plus optional AI interpretation of the resulting table.

When This Analysis Is Useful

Use case	How balancing helps
Year-to-year matrix comparison	Treat one year as starting and another year as target. Coefficients show how categories had to change to explain development.
Budget allocation	Adjust historical departmental or regional spend to a requested row/column budget structure.
Market movement	Compare products, channels, regions, or customers across periods and isolate meaningful cell-level changes.
Forecast reconciliation	Force detailed cells to agree with higher-level forecast totals while preserving observed proportions.
Scenario modeling	Use manually entered targets to ask what detailed data would look like if control totals changed.
True multidimensional reconciliation	Use 4a-4c when multiple selected dimensions must be balanced together, such as Region, Product, Channel, and Department.
Data quality review	Unexpected extreme coefficients can point to missing categories, zeros, outliers, or inconsistent targets.

Implementation Notes From The Code

- The page gets report title and report id from Session and expects Session("dv3") to contain the active report data.
- Scenario controls are shown/hidden in DropDownListScenarios_SelectedIndexChanged. The scenario text in the page explains the expected selections.
- Scenarios 4a-4c call multidimensional table/sum helpers such as MakeMultidimensionalTables and MakeMultidimensionalSumsFromDataTable.
- The main calculations call helpers in App_Code/mFunctions.vb, including BalanceMatrix, CanBalanceMatrixToColRowSums, CanTargetMatrixBalancedFromStartingOne, AdjustMatrixByOverallSum, and DifferencesOfTwoMatrices.
- The grids are stored in Session as GridView1DataSource through GridView6DataSource for export and AI actions.
- The implementation uses small positive replacements for zero matrix cells to avoid divide-by-zero during coefficient calculation.
- Target totals are sometimes adjusted to the starting grand total before balancing, then restored or scaled according to the Adjust by starting values option.

Quick Checklist For Users

- Do you need a row/column matrix scenario (1a-3c), or true multidimensional balancing (4a-4c)?
- Do the selected row/column fields or selected dimensions create a meaningful structure?
- Is Field1 the measure you want to preserve structurally?

- Is Field2 a true target measure, or is it a filter field for start/target values?
- Do manual sums match the number of rows/columns or selected dimensions?
- Are target and starting totals comparable, or should the result be adjusted back to starting totals?
- Do coefficient differences make business sense, or do they reveal missing/incorrect source data?